

Ariun-Erdene Tumurchuluun

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Summary

M.Sc. student in Language Science and Technology at Saarland University and Charles University. Proven expertise in computational linguistics, mechanistic interpretability, and deep learning (PyTorch, TensorFlow, HuggingFace). Fluent in English. Seeking NLP research internship/job in multimodal/multilingual generative AI.

Education

Saarland University | Saarbrücken, Germany

M.Sc. Language Science and Technology

- Expected graduation date in 10/2025. Part of Erasmus Mundus Joint Master's Program (2nd year).
- Coursework: Multimodal Language Understanding, Computational Psycholinguistics, Speech Science, Recent Advances in Mechanistic Interpretability.
- Submitted a short paper to ACL in 2025.

Charles University | Prague, Czech Republic

M.Sc. Language Technologies and Computational Linguistics

- Expected graduation date in 10/2025. Part of Erasmus Mundus Joint Master's Program (1st year)
- Coursework: Deep Learning, Deep Reinforcement Learning, Data Structures, NLP Applications, Complexity and Computability, General Linguistics, Morphological and Syntactic Analysis, Language Data Resources, Speech Recognition and Generation, Unsupervised ML in NLP, Statistical Methods in NLP

National University of Mongolia | Ulaanbaatar, Mongolia

Bachelor degree in Software Engineering | 06/2022

Graduated cum laude and with Honours. Average grade: 3.8/4.0 (top 1%)

Skills

ML/AI/NLP, Mechanistic Interpretability, Docker, Compute clusters, Time Management, Communication skills, Leadership and Teaching, Multimodal LMs, Deep Learning, Speech Technology, Linux/Unix shell, PyTorch, TensorFlow, HuggingFace, Transformers, Docker, Deep reinforcement learning (gaming environments), Knowledge graphs, Sparse Autoencoders, Pyvene, Git

Research & Projects

Causal Investigation of Tense Encoding in Multilingual Transformers: A Thesis Project at Charles University and Saarland University. Supervisors: Prof. Dr. Josef van Genabith, David Mareček, Koel Dutta Chowdhury

Analyzing Word Order Encoding in Multilingual BERT through Probing: Under blind review at ACL (05/2025).

Experience

AI Lab LLC | Ulaanbaatar, Mongolia

Full Stack Developer | 06/2021 - 07/2023

Launched an investment web application. (PHP, Centrifuge, Vue JS, AWS), and a funding web application. (Python, Centrifuge, Docker, Vue JS, AWS); Collaborated with cross - functional teams, defined requirements, and managed sprints via Jira; integrated R&D approach

Empasoft LLC | Ulaanbaatar, Mongolia

Teaching Assistant | 06/2020 - 12/2020

Taught students web development fundamentals (HTML/CSS/JavaScript) and Java programming in Linux lab environments; Designed curriculum and assessments; provided feedback to optimize student learning outcomes.